

Progress Report

Depth of Field Research

Rui Gao — Sep 6

Weeks 1–2 · Aug 24 – Sep 6, 2026

Progress

- Studied how a camera forms an image: the pinhole model, the thin-lens model, focal length, aperture and f-number, and focus distance.
- Learned where the depth-of-field effect comes from: the circle of confusion, and how it grows with aperture size and with distance from the focus plane.
- Studied the basics of rendering and how the computer-graphics pipeline turns 3D geometry into pixels.
- Set up the C++17 / OpenGL 3.3 Core development environment on macOS (CMake, GLFW, GLM, GLAD, Conda).
- Rendered the first triangle and tested the build, run, and screenshot workflow.

Meeting — Aug 24

Attendees: Rui Gao, Beibarys Myrzash, Professor Lingqi Yan.

- First meeting: decided on the project and the general working plan.
- Received the first tasks.

Next Steps

- Work through the graphics pipeline step by step: vertex data, shaders, transformations, and 3D.
- Learn how the depth buffer works, since depth drives the DoF effect.